UPI Transaction Analytics & Fraud Monitoring System

Executive Summary

The Unified Payments Interface (UPI) has transformed digital payments in India by enabling secure, real-time transactions between individuals and businesses. As transaction volumes continue to grow, payment platforms face challenges related to transaction failures, fraud detection, customer satisfaction, and operational efficiency.

This project analyzes over 100,000 UPI transactions along with customer, merchant, device, fraud, and feedback datasets to identify transaction patterns, fraud risks, customer behavior, and operational performance. The project combines Python-based exploratory and statistical analysis with interactive Power BI dashboards to generate actionable business insights.

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1. Business Problem

Digital payment providers require continuous monitoring of:

- Transaction success and failure rates
- Fraudulent transaction patterns
- Merchant and customer risk behavior
- Device-based vulnerabilities
- Customer satisfaction and complaints

The objective is to develop a data-driven decision support system that helps management:

- Monitor transaction performance
- Detect fraud risks
- Improve customer experience
- Optimize operational efficiency
- Enhance platform security

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2. Project Objectives

The project was designed to:

Transaction Analytics

- Analyze transaction volume and value
- Study transaction type preferences
- Measure success and failure rates

Fraud Analytics

- Identify fraud-prone regions
- Detect risky devices and channels
- Monitor suspicious customer behavior

Customer Analytics

- Understand customer engagement
- Analyze satisfaction trends
- Evaluate complaint categories

Merchant Analytics

- Assess merchant risk profiles
- Monitor merchant transaction activity

Dashboard Development

- Build executive dashboards
- Create fraud monitoring reports
- Enable business decision-making through KPIs

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3. Dataset Overview

Seven datasets were integrated into a relational analytical model.

Dataset	Purpose
Customer Master	Customer demographics and risk information
UPI Transactions	Transaction details and status
UPI Account Details	Bank and account information
Merchant Information	Merchant profile and risk scores
Device Information	Device usage and security data
Fraud Alert History	Fraud incidents and resolutions
Customer Feedback Surveys	Satisfaction and issue tracking

Total Records Processed:

- Customers: 10,000+
- Transactions: 100,000+
- Merchants: 500+
- Fraud Alerts: 2,000+
- Feedback Records: 4,000+

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4. Data Cleaning & Preparation

The following preprocessing activities were performed:

Data Quality Checks

- Missing value identification
- Duplicate record removal
- Data type standardization
- Foreign key validation

Feature Engineering

Created analytical metrics such as:

- Transaction Value per Customer

- Fraud Rate
- Transaction Failure Rate
- Merchant Fraud Ratio
- Device Fraud Ratio

Integration

Datasets were merged using:

- customer_id
- transaction_id
- merchant_id
- device_id
- upi_id

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5. Exploratory Data Analysis

Transaction Distribution

Analysis revealed:

Transaction Type	Share
Send	35%
Receive	34.8%
Merchant Payment	25.1%
Bill Pay	5.1%

Send and Receive transactions account for nearly 70% of total platform activity.

Transaction Amount Analysis

Boxplot analysis showed:

- Average transaction value around ₹41.5
- Moderate variability
- Presence of high-value outliers
- Majority of transactions concentrated in lower ranges

These outliers may require additional fraud monitoring.

Device Usage Analysis

Transactions were distributed across:

- Android
- iOS
- Tablet
- Feature Phone

Transaction values remained relatively consistent across device categories, indicating broad platform adoption.

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6. Statistical Analysis

Fraud Rate by Channel

Channel-wise fraud analysis showed:

Channel	Fraud Risk
QR Code	Highest
App	Moderate
Intent	Lowest

QR-based transactions exhibit slightly elevated fraud rates and require enhanced monitoring.

Merchant Risk Distribution

Merchant risk scores were concentrated in low-to-medium ranges.

Observations:

- Most merchants are low-risk

- Few merchants exhibit elevated risk levels
- High-risk merchants require stricter onboarding and monitoring

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Fraud Rate by Hour

Hourly fraud trend analysis showed fluctuations throughout the day.

Peak fraud activity appeared during:

- Late evening
- Night hours

This suggests that fraud monitoring systems should increase alert sensitivity during these periods.

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7. Customer Satisfaction Analysis

Average satisfaction score:

3.61 / 5

Dashboard results indicate generally positive customer sentiment.

Satisfaction Distribution

Majority of responses were:

- Rating 4
- Rating 5

This indicates overall platform acceptance and positive customer experience.

Issue Analysis

Most common customer complaints:

1. Transaction Issues
2. App Usability Issues

3. Fraud Concerns
4. Other Issues

Transaction-related complaints dominate feedback records, indicating a need to improve transaction reliability.

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8. Dashboard Insights

Executive Dashboard

Key KPIs:

KPI	Value
Total Transactions	100K
Transaction Failure Rate	5.87%
Total Transaction Amount	₹4.15M
Fraud Rate	2.00%
Average Transaction Value	₹41.53
Customer Satisfaction	3.61

Regional Analysis

Transaction values were distributed fairly evenly across:

- North
- East
- West
- Central
- South

This demonstrates balanced adoption of UPI services across regions.

Device Analysis

Transaction value contribution by device:

- Feature Phone
- Android
- iOS
- Tablet

All devices contributed nearly equal transaction volumes, highlighting widespread accessibility.

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9. Fraud Monitoring Dashboard Findings

Fraud dashboard revealed:

Key Risk Indicators

Metric	Value
Fraud Rate	2.00%
Fraud Alerts	2,000
Failure Rate	5.87%
High Risk Customers	1,408

Major Fraud Categories

Most frequent alerts:

- Unusual Time Activity
- Frequent Failures
- Suspicious Login
- Unusual Transaction Amount

These categories represent the primary sources of fraud risk.

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10. Business Recommendations

Recommendation 1

Strengthen QR Transaction Security

Implement:

- Dynamic QR verification
- Behavioral monitoring
- Risk-based authentication

Expected Outcome:

- Reduction in fraud incidents
- Improved transaction trust

Recommendation 2

Enhance Failure Detection

Focus on:

- Network-related failures
- Incorrect PIN attempts
- Bank downtime events

Expected Outcome:

- Lower transaction failure rate
- Better customer experience

Recommendation 3

Monitor High-Risk Customers

Create:

- Customer risk segmentation
- Automated fraud alerts
- Enhanced verification workflows

Expected Outcome:

- Reduced fraud exposure
- Faster incident response

Recommendation 4

Improve Customer Support

Prioritize:

- Transaction complaints
- App usability concerns
- Fraud-related assistance

Expected Outcome:

- Higher satisfaction scores
- Increased customer retention

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11. Technologies Used

Technology	Purpose
Python	Data Cleaning & Analysis
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib	Visualization
Seaborn	Statistical Visualization
SQL	Data Storage & Querying
Power BI	Dashboard Development
GitHub	Project Hosting

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12. Conclusion

This project successfully developed an end-to-end UPI analytics framework capable of monitoring transaction performance, customer behavior, fraud patterns, merchant risk, and customer satisfaction. Through Python-based analytics and Power BI dashboards, the solution provides management with actionable insights for improving operational efficiency, reducing fraud, and enhancing user experience.

The findings indicate a healthy transaction ecosystem with strong customer adoption, moderate fraud exposure, and opportunities for improvement in transaction reliability and fraud prevention. The developed dashboards enable real-time monitoring and support data-driven decision-making for digital payment platforms.

This report is strong enough to include directly in your GitHub repository as `Project_Report.md` or convert into a PDF for your portfolio.